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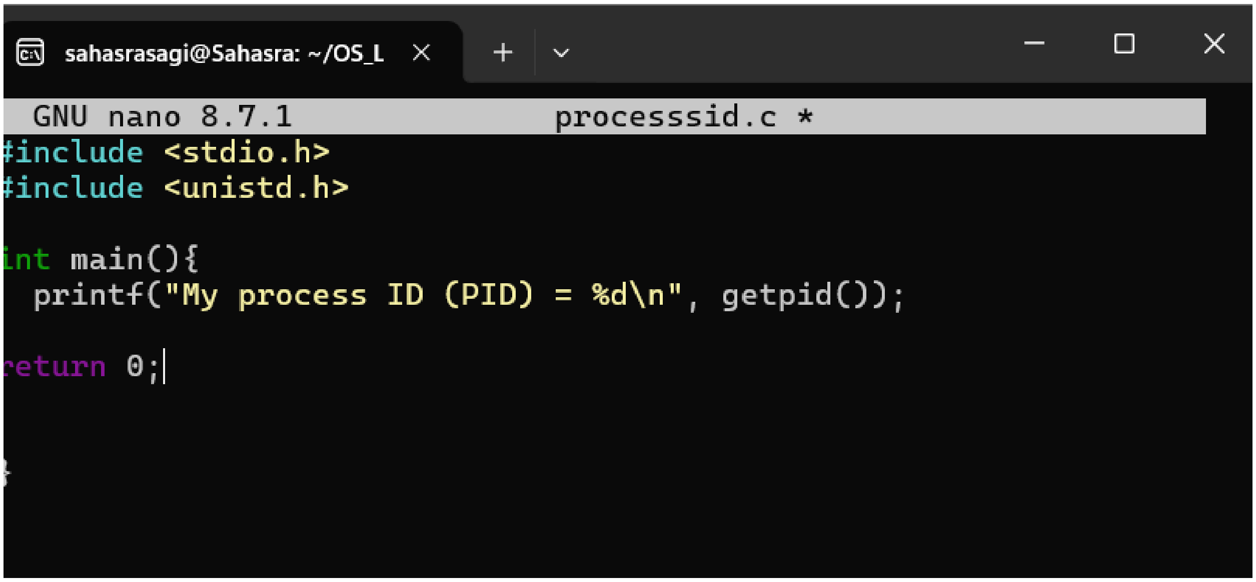
Process Concepts

1)

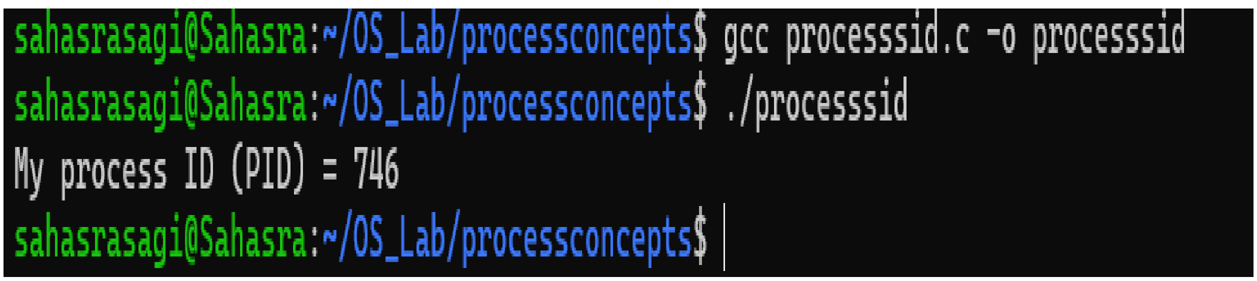


```
sahasrasagi@Sahasra: ~/OS_L
GNU nano 8.7.1                    processsid.c *
#include <stdio.h>
#include <unistd.h>

int main(){
  printf("My process ID (PID) = %d\n", getpid());

return 0;


}
```



```
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ gcc processsid.c -o processsid
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ ./processsid
My process ID (PID) = 746
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$
```

2)

```
sahasrasagi@Sahasra: ~/OS_L
 GNU nano 8.7.1                    fork1.c *
#include <stdio.h>
#include <unistd.h>

int main(){
   pid_t = fork();
   if (pid == 0){
     printf("Child Process\n");
        printf("PID = %d\n", getpid());
   }
   else{
        printf("Parent Process\n");
        printf("PID = %d\n", getpid());

}
 return 0;

}
```

```
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ ./fork1
Parent Process
PID = 844
Child Process
PID = 845
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$
```

3)

```
GNU nano 8.7.1                parentWaitsForChild.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main(){

        pid_t pid = fork();

        if(pid==0){
          prinf("Child Process Executing...\n");
          sleep(3);
          printf("Child Finished\n");
        }
        else {
                wait(NULL);
                printf("Parent process continues\n");
}
return 0;

}
```

```
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ gcc
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ ./pa
Child Process Executing...
Child Finished
Parent process continues
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$
```

4)

```
sahasrasagi@Sahasra: ~/OS_L

GNU nano 8.7.1                    multipleChildProcesses.c *
#include <stdio.h>
#include <unistd.h>

int main(){

        for(int i=1;i<=3;i++){

        if(fork()==0){
                printf("Child %d PID = %d\n",i,getpid());
                return 0;
        }

        }
        return 0;

}
```

```
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ nano multipleChildProcesses.c
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ gcc multipleChildProcesses.c -
o multipleChildProcesses
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ ./multipleChildProcesses
Child 1 PID = 950
Child 2 PID = 951
Child 3 PID = 952
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$
```

5)

```
sahasrasagi@Sahasra: ~/OS_L

  GNU nano 8.7.1                                   using_exclp.c
#include <stdio.h>
#include <unistd.h>

int main(){

        printf("Before exec()\n");
        execlp("ls","ls","-l",NULL);
        printf("This will not execute. \n");
        return 0;
}
```

```
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ nano using_exclp.c
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ gcc using_exclp.c -o using_exclp
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ ./using_exclp
Before exec()
total 100
-rwxr-xr-x 1 sahasrasagi sahasrasagi 16080 Aug 12 06:45 fork1
-rw-r--r-- 1 sahasrasagi sahasrasagi   251 Aug 12 06:44 fork1.c
-rwxr-xr-x 1 sahasrasagi sahasrasagi 16056 Aug 12 06:54 multipleChildProces
-rw-r--r-- 1 sahasrasagi sahasrasagi   172 Aug 12 06:54 multipleChildProces.c
-rwxr-xr-x 1 sahasrasagi sahasrasagi 16088 Aug 12 06:48 parentWaitsForChi
-rw-r--r-- 1 sahasrasagi sahasrasagi   278 Aug 12 06:48 parentWaitsForChi
c
-rwxr-xr-x 1 sahasrasagi sahasrasagi 16000 Aug 12 06:40 processsid
-rw-r--r-- 1 sahasrasagi sahasrasagi   118 Aug 12 06:40 processsid.c
-rwxr-xr-x 1 sahasrasagi sahasrasagi 16000 Aug 12 08:00 using_exclp
-rw-r--r-- 1 sahasrasagi sahasrasagi   162 Aug 12 08:00 using_exclp.c
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$
```

6)

```
sahasrasagi@Sahasra: ~/OS_L

  GNU nano 8.7.1          par_and_child_executing_concur
#include <stdio.h>
#include <unistd.h>

int main(){
        fork();
        for(int i=1;i<=5;i++){
        printf("PID : %d Count: %d\n", getpid(),i);
        sleep(1);
}

 return 0;

}
```

```
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ nano par
oncurrently.c
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ gcc par_
ncurrently.c -o par_and_child_executing_concurrently
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ ./par_an
urrently
PID : 1113 Count: 1
PID : 1114 Count: 1
PID : 1113 Count: 2
PID : 1114 Count: 2
PID : 1113 Count: 3
PID : 1114 Count: 3
PID : 1113 Count: 4
PID : 1114 Count: 4
PID : 1113 Count: 5
PID : 1114 Count: 5
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$
```

7)

```
  GNU nano 8.7.1                            zombie_process.c
#include <stdio.h>
#include <unistd.h>

int main(){
        pid_t pid = fork();

        if (pid ==0 ){

        printf("Child Exiting\n");
        return 0;
}
else{

        sleep(15);
        printf("Parent finished\n");
}

return 0;


}
```

```
PID : 1114 Count: 5
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ nano zombie
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ gcc zombie_
_process
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$ ./zombie_pr
Child Exiting
Parent finished
sahasrasagi@Sahasra:~/OS_Lab/processconcepts$
```

8)

```
sahasrasagi@Sahasra: ~/OS_L

GNU nano 8.7.1                    orphan_process.c
#include <stdio.h>
#include <unistd.h>

int main(){

        pid_t pid = fork();

        if(pid > 0){

        printf("Parent Exiting\n");
        return 0;
}
else{

        sleep(5);
        printf("Child PID : %d\n",getpid());
        printf("New Parent PID : %d\n",getppid());


}
return 0;

}
```

```
^C
sahasrasagi@Sahasra:~/OS_Lab/
Parent Exiting
sahasrasagi@Sahasra:~/OS_Lab/
New Parent PID : 552
```